Building quantum codes with transversal CCZ

from projective geometry

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Joint work with Kenneth R. Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

A code can protect information and implement a gate

The construction

48 physical qubits encode 3 logical qubits with distance 3. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ.

- A *logical qubit* is one qubit encoded across a larger block.
- Distance 3 permits correction of an arbitrary single-qubit error.
- A *transversal* layer acts separately on each physical qubit, so one faulty gate affects one qubit.
- T (one qubit) and CCZ (three qubits) are non-Clifford gates, defined on slide 5. Clifford gates alone are not universal.

The code must satisfy both error-correction constraints and gate constraints.

Binary codes describe a quantum encoding

Let S be a full-rank matrix of X-type checks (stabilizer generators) and K a matrix of k logical rows.

$$\mathcal{C}_2 = \mathrm{rowspan}(S) \subset \mathcal{C}_1 = \mathrm{rowspan} \begin{bmatrix} K \\ S \end{bmatrix}.$$

A CSS code [CS96, Ste96] has the orthonormal logical basis

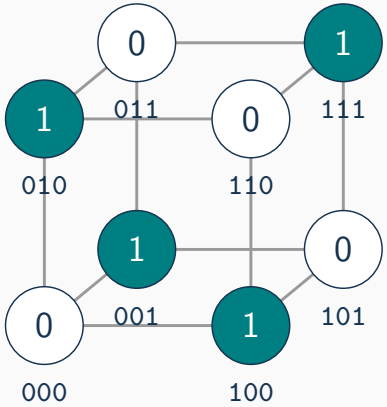
$$|x\rangle_L = \frac{1}{\sqrt{|\mathcal{C}_2|}} \sum_{v \in \mathcal{C}_2} |xK \oplus v\rangle, \qquad x \in \mathbb{F}_2^k.$$

Here $|c\rangle$ is the basis state labelled by the binary word c. Rows of K are independent modulo $\mathcal{C}_2$. X flips a physical bit. Z changes its sign when that bit is 1. The X checks come from $\mathcal{C}_2$. The Z checks come from $\mathcal{C}_1^\perp$.

Logical basis states are equal superpositions over distinct classical cosets.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

The eight-qubit cube starts with affine functions



Vertex values: truth table of $u_1 \oplus u_2 \oplus u_3$.

Physical positions: $u \in \mathbb{F}_2^3$.

$$f_{x,b}(u) = b \oplus x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3.$$

A truth table lists the 8 values of a function on the cube. The truth tables of the $f_{x,b}$ form $\mathrm{RM}(1,3)$, an $[8,4,4]$ binary code.

Take $S = (11111111)$, the truth table of the constant 1, and $K = (u_1, u_2, u_3)^\top$.

The same cube supplies physical positions, generator rows, and a gate pattern.

Complementary words encode three logical qubits

The classical code has 16 words, partitioned into 8 complementary pairs:

$$|x\rangle_L = \frac{|f_{x,0}\rangle + |f_{x,1}\rangle}{\sqrt{2}}, \qquad x \in \mathbb{F}_2^3.$$

For example,

$$|000\rangle_L = \frac{|00000000\rangle + |11111111\rangle}{\sqrt{2}}.$$

The eight disjoint pairs give eight orthogonal states: $k = 3$ logical qubits.

Flipping all 8 bits exchanges the two words and leaves the encoded state unchanged.

A diagonal gate must assign the same phase to both words of each pair.

Single-qubit phases can produce a three-qubit CCZ

$$T = \mathrm{diag}(1, e^{i\pi/4}), \qquad T^\dagger = T^{-1}.$$

Logical CCZ multiplies $|111\rangle_L$ by -1 and leaves the other 7 logical labels unchanged:

$$\mathrm{CCZ}|x\rangle_L = (-1)^{x_1 x_2 x_3}|x\rangle_L.$$

Choose $T^\dagger$ at parity-one vertices $p(u) = u_1 \oplus u_2 \oplus u_3$ and T elsewhere. A basis word c gains $e^{i\pi/4}$ for each 1 at a T site and $e^{-i\pi/4}$ for each 1 at a $T^\dagger$ site:

$$W(c) = \sum_u (-1)^{p(u)} c(u), \qquad \text{phase on } |c\rangle = e^{i\pi W(c)/4}.$$

The toy gate works, but its distance is two

Each site contributes $(p \oplus c)(u) - p(u)$ to W, and $\mathrm{wt}(p) = 4$, so $W(c) = d_H(p, c) - 4$. For $x \neq 111$, $p \oplus f_{x,b}$ is nonconstant affine and has weight 4:

logical label	$W(f_{x,0}),\ W(f_{x,1})$	phase
$x \neq 111$	$0,\ 0$	$+1$
$x = 111$	$-4,\ +4$	-1

Both words in each pair acquire the required phase.

The error $Z_{000}Z_{100}$ commutes with the all-X check and acts as $(-1)^{x_1}$ on $|x\rangle_L$. Thus the cube code [CH17] has parameters $[\![8, 3, 2]\!]$.

The 48-qubit construction keeps the signed-weight phase rule and raises the distance to 3.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Overlap counts control weights of sums

For binary words a, b, their product $a \wedge b$ selects the common ones.

$$\mathrm{wt}(a \oplus b) = \mathrm{wt}(a) + \mathrm{wt}(b) - 2\,\mathrm{wt}(a \wedge b).$$

For three words the inclusion–exclusion coefficients are $1, -2, 4$:

$$\begin{aligned}\mathrm{wt}(a \oplus b \oplus c) = {} & \mathrm{wt}(a) + \mathrm{wt}(b) + \mathrm{wt}(c) \\ & - 2\big(\mathrm{wt}(a \wedge b) + \mathrm{wt}(a \wedge c) + \mathrm{wt}(b \wedge c)\big) \\ & + 4\,\mathrm{wt}(a \wedge b \wedge c).\end{aligned}$$

The identity holds for ordinary or signed weights. Terms of order 4 vanish modulo 8.

The general gate condition is a signed-weight congruence

Write $\Gamma_j = +1$ for T and -1 for $T^\dagger$, and define

$$W_\Gamma(c) = \sum_{j=1}^{n} \Gamma_j c_j.$$

A physical gate layer implements logical CCZ exactly when

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$$

for every $x \in \mathbb{F}_2^3$ and $y \in \mathbb{F}_2^s$. The modulus is 8 because $e^{i\pi W/4}$ depends only on W mod 8.

Dependence on x produces the logical gate. Independence from y preserves each stabilizer coset.

A gate-design question becomes an integer congruence on a binary code.

Only single, pair, and triple overlaps are needed

Expand $W_\Gamma(xK \oplus yS)$ with slide 7 and match the coefficients $1, -2, 4$ against $4x_1x_2x_3$. For distinct rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
$$W_\Gamma(g_a \wedge g_b) \equiv 0 \pmod 4,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv \begin{cases} 1 \pmod 2, & \{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}, \\ 0 \pmod 2, & \text{otherwise.} \end{cases}$$

Γ_j is odd, so signed and unsigned triple weights agree mod 2. The unsigned parities are the nine *CH conditions* [CH17], which allow a diagonal Clifford correction: *quasi-transversal* CCZ. The signed congruences characterize a physical $T/T^\dagger$ layer with no correction.

All nine parity conditions

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

From magic-state distillation to triorthogonal codes

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices give CSS codes with logical T, up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via unital spaces and RM indicators.

Our CCZ needs an odd logical triple. Ordinary triorthogonal matrices have even distinct-row triples.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

A triorthogonal matrix and a triorthogonal space differ

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows label logical T outputs. Even-weight rows give X checks.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j = 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every word in $\mathcal{H}$ has even weight.

Nezami–Haah classify parent spaces, then obtain codes by puncturing and deleting coordinates.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.
[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Geometry fixes the stabilizers before the search

1. Choose distinct nonzero columns for S: guarantee $d_Z \geq 3$.
2. Choose an 8-divisible stabilizer code: every stabilizer weight is a multiple of 8.
3. Search the logical rows K, then test physical signs Γ.

8-divisibility enforces the stabilizer-only overlap parities: by slide 7, weights $\equiv 0$ mod 8 force pair overlaps $\equiv 0$ mod 4 and triple overlaps even. The mixed and logical-row conditions remain to be solved.

The SAT search has no distance constraint and does not search over S.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \quad \Longrightarrow \quad \text{every weight-one error is detected,}$$

$$\beta_i \neq \beta_j \quad \Longrightarrow \quad \beta_i \oplus \beta_j \neq 0.$$

Thus distinct nonzero columns detect every weight-two Z error.

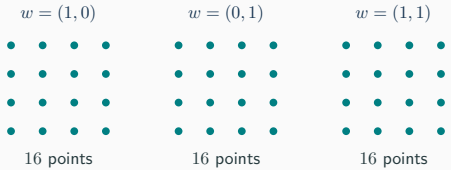
Over $\mathbb{F}_2$, those columns are points of the projective space $\mathrm{PG}(s-1,2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the stabilizer support, independently of the logical labels.

The support is projective space minus a subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the 15 nonzero points with $w = 00$, so exactly the points with $w \neq 00$ remain.



The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ generate S.

Three cosets, each with 16 positions, give a projective $[48, 6]$ stabilizer code.

Hyperplane sections give the two nonzero weights

A stabilizer word is the evaluation of

$$\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w.$$

- $\lambda \neq 0$: eight ones in each of the 3 blocks, weight $3 \cdot 8 = 24$.
- $\lambda = 0$, $\mu \neq 0$: one on 2 of the 3 blocks, weight $2 \cdot 16 = 32$.

$$\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|.$$

Projective two-weight codes and their geometry: [CK86].

Weights 24 and 32 are multiples of 8. Geometry supplies the divisibility.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

SAT assigns three logical bits to each geometric point

Unknowns: the column labels $\alpha_j \in \mathbb{F}_2^3$ of K.

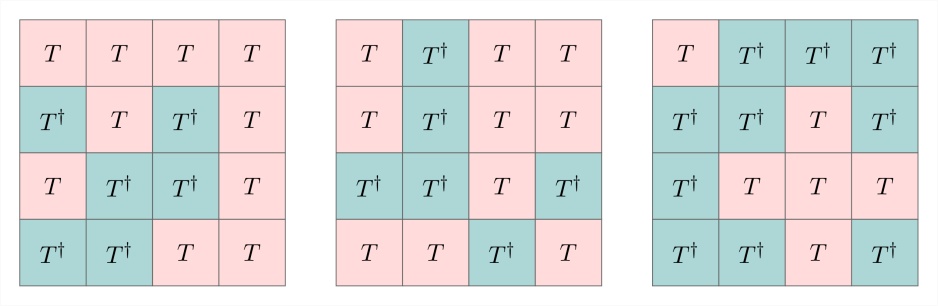
$$3n = 144 \text{ label bits}, \qquad 7 + 6s + 3\binom{s}{2} = 88 \text{ parity constraints (not CNF clauses).}$$

K1–K3 give 7, KS1–KS2 give $6s$, and KS3 gives $3\binom{s}{2}$, for 88 at $s = 6$.

- Stabilizer-only conditions already hold.
- Distance constraints are absent because S guarantees $d_Z \geq 3$.
- Remaining parity conditions include products of up to three label bits.
- A compatible labeling is followed by a separate physical-sign feasibility test.

Explicit logical rows

The published pattern is a layer of 48 phase gates



$26\,T + 22\,T^\dagger \implies$ logical CCZ on 3 encoded qubits.

Pink: T. Teal: $T^\dagger$. Three 16-qubit blocks. Pattern from [LB26].

Pattern formula

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

A three-point witness makes the Z-distance exact

Put a Z error at $u = 0000$ in each of the 3 blocks. The u parts of the three columns are 0000.

$$\underbrace{10 \oplus 01 \oplus 11}_{\text{stabilizer syndrome}} = 00, \qquad \underbrace{011 \oplus 110 \oplus 000}_{\text{logical label sum}} = 101 \neq 000.$$

The operator commutes with every check but acts nontrivially on encoded information.

$$Q_{48} = [\![48, 3, 3]\!], \qquad (d_X, d_Z) = (16, 3).$$

Projectivity proves $d_Z \geq 3$. The explicit witness proves $d_Z \leq 3$.

The lower bound turns supports into Reed–Muller words

Nezami–Haah [NH22] encode column supports by Boolean indicator polynomials and classify their low-weight Reed–Muller words.

For our projective stabilizer support $\mathcal{P} \subset \mathbb{F}_2^s$, let $\chi_{\mathcal{P}}$ be its indicator. Let c be $\chi_{\mathcal{P}}$ or $\chi_{\mathcal{P}}$ plus the indicator of the origin, whichever has even weight. The stabilizer-overlap parities give

$$c \in \mathrm{RM}(s-4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

A short code must therefore arise from a low-weight Reed–Muller word, followed by compatible logical labels.

Indicator correspondence

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Concurrent classification closes the 39–46 gap

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$ only	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$, up to CNOT+S output equivalence. The nine CH conditions select their CCZ output class.

Their classification does not test physical-sign feasibility for every code.

What is proved at 47?

[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).
[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

The geometric construction produces a family

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

Each support requires compatible logical rows and physical signs. Existence is not automatic from the point count.

The next algebraic task is to characterize which supports admit the required completions.

Gate-aware search complements general SAT code design

DalFavero et al. [DWL+26]

- Discover or extend stabilizer codes.
- Prescribe detectable or correctable errors.
- Incorporate symmetries and hardware noise.

This construction

- Fix geometric stabilizers first.
- Search the logical rows for CCZ.
- Test the physical gate signs.

Error-detection constraints alone do not guarantee fault-tolerant logical operations.

Code length and circuit footprint

[DWL+26] B. DalFavero et al., Satisfying quantum codes: physics-informed and hardware-aware code design with SAT solvers, arXiv:2608.23460 (2026).

Open questions connect algebra to implementation

- Characterize geometric supports admitting compatible logical rows and physical signs.
- Construct higher-distance families and extensions to other non-Clifford gates or more logical qubits.
- Determine the minimum length for physical $T/T^\dagger$ CCZ without a correction.
- Compare full circuits, noise, and resource costs before claiming a practical advantage.

Paper
arXiv:2610.09341

Code and data
github.com/
jerrylvx/
gate2code-public

Geometry supplies distance and divisibility. Logical labels and signed weights supply the gate.

Joint work with Kenneth R. Brown. Thanks to Robert Calderbank for classical coding-theory pointers.

The nine parity conditions

K1	$\mathrm{wt}(K_i) = 0 \pmod 2$
K2	$\mathrm{wt}(K_i \wedge K_j) = 0 \pmod 2$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) = 1 \pmod 2$
KS1	$\mathrm{wt}(K_i \wedge S_l) = 0 \pmod 2$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) = 0 \pmod 2$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) = 0 \pmod 2$
S1	$\mathrm{wt}(S_l) = 0 \pmod 2$
S2	$\mathrm{wt}(S_l \wedge S_m) = 0 \pmod 2$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) = 0 \pmod 2$

Distinct logical indices $i < j$ and stabilizer indices $l < m < r$.

K1–K3 select CCZ: in the target phase $4x_1x_2x_3$, only the coefficient of $x_1x_2x_3$ is nonzero.

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A logical row is a function on each block

The table specifies K_1:

w	10	01	11
$K_1(u, w)$	$u_1 \oplus u_3$	$1 \oplus u_3$	u_4

K_2 and K_3 are quadratic on each block [LB26]. Evaluating the functions at all 16 addresses gives each block of the row.

The three rows must be independent modulo $\mathcal{C}_2$ and satisfy the overlap parities. A candidate labeling still needs a feasible physical-sign pattern. Full matrices and independent verification are in the public repository.

Return to main talk

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

The physical signs have a polynomial formula

Set $\Gamma_j = 1 - 2\gamma_j$, where $\gamma_j \in \{0, 1\}$ marks $T^\dagger$.

$$\gamma \cdot c \equiv \frac{\mathrm{wt}(c)}{2} - 2x_1x_2x_3 \pmod 4, \qquad c = xK \oplus yS.$$

The unknowns remain binary even though the congruence is modulo 4.

$$\begin{aligned} \gamma(u, w) &= u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u), \\ Q_1 &= u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4, \\ Q_2 &= u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4. \end{aligned}$$

The resulting 22 flips satisfy the phase condition on all $8 \times 64 = 512$ codewords.

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Why a Reed–Muller indicator appears

Over $\mathbb{F}_2$, the stabilizer conditions say

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0, \qquad 1 \le t \le 3.$$

Let $c = \chi_{\mathcal{P}} + \epsilon\delta_0$, choosing ϵ so $\mathrm{wt}(c)$ is even. The origin changes no positive-degree moment. Then $c \perp \mathrm{RM}(3, s)$, hence

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

Rank reduction and low-weight Reed–Muller classifications [NH22] make the $n \le 38$ search finite.

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[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

What the 47-qubit comparison establishes

Jacinto et al. [JVB+26] give a $[\![47, 3, 3]\!]$ code satisfying the nine CH conditions. Its physical $T/T^\dagger$ sign system is infeasible [LB26].

Wills–Jain–Singh [WJS26] establish minimum length 47 in the quasi-transversal CCZ class with $k = 3$, $d_Z \geq 3$. Their catalogue identifies outputs up to CNOT and S, not full Clifford equivalence.

These facts do not establish uniqueness of the length-47 code or rule out every other length-47 completion with feasible signs.

Return to main talk

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).
[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Code length and distillation footprint are different resources

Apply the phase gates to encoded $|+\rangle^{\otimes 3}$, measure checks, and decode to prepare a CCZ resource state.

- n: code length, or number of noisy T resource inputs in the distillation description.
- N: circuit footprint in the distillation implementation of [JVB+26]. For the matrix description, $N = k + s$ [WJS26].
- The 47-to-1 distance-3 CCZ protocol has $(n, N) = (47, 9)$.

At least 3 input Z errors are needed for an undetected logical error under the ideal-Clifford distillation model. A complete physical factory additionally needs a circuit, preparation, measurements, decoding, and a noise model.

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[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).